Course Notes — SESSION 6
KNOWLEDGE REPRESENTATION AND REASONING

1. Introduction

Our previous session proposed learning to assign credit by representing problems as **networks of constraints**, and **propagating** information across these networks. In this session, we will build on these proposals by studying how AI systems and organizations may: (1) augment constraints with representations of *knowledge*, or what is often called in AI a *knowledge base*; (2) augment basic ways of evaluating constraints with more complex *reasoning* strategies, or what is often called in AI an *inference engine*. We will call all such AI ideas as strategies for **knowledge representation and reasoning**, or just "KRR".

To make a comparison, consider how problem-solving in the AI idea of search was based on learning to use heuristics (educated guesses) to select actions. In this **heuristic approach**, it was assumed that we *already knew the problem*: the search tree was given. In fact, even in our example from the previous session of Sudoku, we could just use a heuristic (for guessing the value of a square) to keep the propagation process going, since it was given that we were playing Sudoku. KRR instead is based on an **epistemological approach**, in which the assumption is that we have to get to know problems before even considering how to search for solutions to them.

Our guiding light for exploring KRR will be mathematician **John McCarthy** (1927-2011), co-founder of the MIT AI lab with Marvin Minsky (and later of the AI lab at Stanford University), who pioneered KRR ideas in AI. He proposed that AI systems can get to know particular problems by representing knowledge of general, stable facts about the world, which he called representations of knowledge about **phenomena**. Knowledge about phenomena can be used to reason about a specific problem-solving environment, which he called a **situation**. Hence, for McCarthy, learning to assign credit to solve problems is a process of *learning to represent and reason about knowledge about phenomena as they relate to real-world situations*.

To illustrate, McCarthy gave the example of a supermarket that wants to use data about shoppers to send them targeted ads based on predicting what they would like to buy. A heuristic approach might simply analyze variables observed in supermarket receipt data (e.g., food items and prices) and data about customers (e.g., names, addresses). Indeed, data science techniques today, including ones using recent LLM tools, are almost all based on such analyses. McCarthy thought all all this is naïve. Variables in the supermarket's data also contain rich facts about the *phenomenon of supermarket shopping*. We all know these facts, yet they are typically ignored.

For example, consider if a shopper comes to the supermarket rarely, and always buys frozen spinach. We can infer facts such as: (1) the shopper prefers the spinach at the supermarket versus where she or he normally shops, and (2) the shopper has a freezer at home. Beyond just targeted ads for frozen spinach or other frozen foods, these facts could be used to reason about the shopper's overall "situation" and how to change it. For example, ads could bundle discounts on shopping baskets over a certain amount of frozen foods to incentivize the shopper to do their regular shopping at the supermarket, rather than just stopping in for spinach.

McCarthy further points out that this epistemological approach offers not only potentially distinctly intelligent solutions to problems but, remarkably, also vast *efficiency* gains. In the supermarket example, one could simply infer ways of targeting ads based on a few basic elements of our knowledge of shopping. In contrast, digital marketing techniques (and data science in general) often rely on vast databases about large numbers of customers, and computationally-intensive prediction techniques just to get blunt insights.

On the other hand, consider how, to change such a shopper's "situation", a marketer might ideally want to observe many variables and infer many facts, and do so continually as shoppers' situations themselves change. For example, the shopper may wish to buy more or less spinach according to how much space is currently in her or his freezer, whether relatives are visiting from out of town, or whether the shopper has just read a news story touting the health benefits of spinach. How could marketing problems, or any problems in AI systems or organizations, be solved intelligently given the vast complexity of knowledge we would need to infer about real-world phenomena and the situations in which they arise?

To answer this question, McCarthy noted that using our knowledge of phenomena to take actions in real-world situations is generally what humans do to get along in the world. This is the case whether it is for sending targeted ads in a supermarket, to our everyday activities such as taking a walk or having a conversation. In McCarthy's parlance, we do all these activities based on our **commonsense knowledge** (all our vast set of default assumptions about how the world generally works) and **commonsense reasoning** (our basic ways of reasoning about real-world situations that we encounter as we take actions). In this view, the ability of an AI system or organization to represent and reason about knowledge about any particular problem (such as targeting supermarket ads to shoppers) depends on (indeed, presupposes) a base of commonsense knowledge and reasoning. Hence, the general problem of credit assignment in AI ultimately reduces down to — or is most fundamentally about — the challenges of representing and reasoning about commonsense knowledge.

2. The Advice taker and the frame problem

What are the possibilities and challenges of representing and reasoning about commonsense knowledge in AI systems and organizations? In 1959, McCarthy outlined a vision of the possibilities for KRR in terms of a hypothetical AI system that he called the "**advice taker**". A decade of research later, in 1969, he outlined what he had found to be the fundamental credit assignment challenge in KRR that he called the "**frame problem**".

2.1 Advice taker: envisioning an AI system that represents and reasons about knowledge

McCarthy's proposed AI system can be summarized as having the following characteristics:

> (1) The Advice Taker easily learns new representations **just by you telling it facts about the situation** ("giving it advice"), rather than by having to be programmed in complex sequences of code. This characteristic implies that anyone should be able to build a customized AI system just by having a conversation with it in natural language (English, Korean, etc.). McCarthy's vision was that AI systems could then become like public utilities (i.e., like running water or electricity): accessible to everyone, rather than controlled by a small number of programmers.
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> (2) The Advice Taker **represents knowledge as concepts and facts** that it can use across situations, rather than as procedures or algorithms designed for specific situations. Think of how a step-by-step recipe for an omelet is hard to modify since changing anything about even one step (such as the temperature of the pan) can easily change the effects of other steps. In contrast, concepts and facts about omelets are more stable, such as the fact that omelets usually have eggs, and the concept that heating eggs in a pan causes the eggs to solidify. McCarthy's vision was for AI that was not bound to rigid instructions, and which instead could flexibly draw on concepts and facts across situations.
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> (3) The Advice Taker **precisely, logically reasons** about concepts and facts to modify representations (and guide actions) across situations. McCarthy's vision was that valid, sound reasoning based on concepts and facts would enable reliable modification of representations (i.e., reliable credit assignment), rather than assigning credit under the reinforcement paradigm or by using search.

McCarthy proposed that such an "Advice Taker" AI system could be generally intelligent when it is provided with (1) enough knowledge (concepts and facts) to *represent* any problem in the world — what he called **commonsense knowledge**), and (2) with the capacity to *reason* efficiently and logically about specific situations by using this general knowledge (what he called **commonsense reasoning**). Hence, McCarthy's overall point of view was that an intelligent AI system or organization is one that has *common sense*, defined as a *general capacity to represent and reason about knowledge across situations*.

2.2 The frame problem: KRR and the credit assignment problem

What are the challenges of achieving his Advice Taker vision? The obvious challenge is that there is presumably a lot of commonsense knowledge about the world, such that representing it is a big endeavor. McCarthy, however, made a not-so-obvious and pioneering insight in this regard. To see his insight consider that, in search, the idea of a heuristic is that, *if* one is state X of a search tree, *then* one should consider action Y. Following this logic, an AI system merely can store knowledge of all the possible states of a problem, and then learn to select actions for each state. McCarthy argues that this belief in relying just on heuristics about *what to do when* is naïve. This is because one *also* has to have knowledge of what *not to do* when — e.g., that W, Z or any other actions do not apply. It implies that representing and reasoning about problems is a vastly more complex challenge that implied in a search approach. Assigning credit depends on understanding how, to borrow a movie metaphor, "frame-by-frame" changes in situations *both* make certain actions relevant while leaving others irrelevant. Following this metaphor, McCarthy called this challenge the "**frame problem**".

To get a sense of the frame problem, imagine that I'm a soccer player, and I have the ball near the goal about to take a shot, while the goalkeeper is standing by the goal waiting to try to stop the ball when I kick it. If I am intelligent, then I need to know that the action of kicking the ball will very possibly have certain outcomes. Obviously, these outcomes could be various, and they're not limited to the motion of the ball. Depending on the situation, there might be consequences for what happens with, say, each of my teammates and the defenders (all of whom might move, though in different ways, e.g., one defender tries to slide in front of me, another one tries to guard my teammate, and so on), with the refs, with the fans who stand up and cheer or boo, and so on, and so on. That's tough. I have lots of things to think about — in the language of this course, I have lots of information to process, or lots of symbols to manipulate.

Yet worse, I also have to represent, or in some other way know, **the things that won't change** from one moment to the next. This is the *frame problem*. Virtually nothing that happens as a result of my throw will affect the position of the goal posts, or the lines of the soccer field, or the location of the stadium, or the height of the outfield walls; it also won't affect the location of the highways leading to the ballpark or any other highways; and it won't affect the weather in Thailand or the price of bread in France or the size of the ancient pyramids in Mexico or the shape of a pine cone in Korea, and so on. All of which (or at least tons of which) an intelligent AI system or organization has to know in order to be intelligent. If you don't know tons of that stuff, you're not intelligent.

Taken together, the above examples of the frame problem suggest that the challenge of credit assignment in AI cannot be solved just by statistical prediction or trial-and-error actions of search, nor by representations of only given problems. Such a system would need to know not just how to assign credit to a set of interdependent actions. It would need to assign credit to the events and actions of the entire universe and do so dynamically as situations unfold frame-by-frame.

One might want to say that the problem is solved with some notion of relevance, i.e., that the price of bread and kicking a soccer ball have no relevance for one another. Yet this seems entirely to beg the question: how could one have knowledge of — or be able to compute — relevance itself? How do we rule things out as irrelevant without already knowing tons and tons about them? For McCarthy, the answer is that firstly, in fact, any intelligent system *does* need to know tons of this stuff, or what he called "commonsense knowledge". Further, it needs to be able to efficiently access and make inferences from this knowledge, called "commonsense reasoning", with a characteristic called "elaboration tolerance".

3. Representing commonsense knowledge

For McCarthy, common sense refers to knowledge that is common to most humans. Common sense can be granular, such as the common sense specific to a local culture or practice. The basic idea of common sense in AI, however, is that it is knowledge that most humans have regardless of their cultural background. This knowledge includes representations of:

(1) Facts about events in the world;
(2) Which actions are possible (by the intelligent system itself, or by other agents) given the facts of a situation;
(3) What the probable effects of these actions will be; and
(4) What the physical world is like and how it is perceived, including the properties of objects and relations of these objects and properties to one another.
(5) Concepts and principles to abstract all of the above knowledge.

3.1 Knowledge representations need to be declarative

Let us introduce a distinction between **explicit** and **implicit knowledge**. Explicit means stored in minds or external memory, such as paper or a computing device. Implicit can be illustrated by the sentence: *"The Pyramids of Giza in Egypt are older than my neighbor's house"*. If I tell you this, you believe me, even though it is almost sure that neither I nor anyone else in history has spoken this sentence. You believe me since there are facts about (1) Egyptian pyramids (they're usually ancient) and (2) neighbors' houses (they're usually not ancient). The connection between these two facts is called a **logical entailment**. This entailment was *implicit* in your commonsense knowledge: *you already "knew" the pyramids are older than my neighbor's house, even if you had never thought about it*. If most commonsense knowledge is implicit, then we can represent it efficiently by representing a still vast, yet much smaller amount of explicit knowledge.

McCarthy proposed that this efficiency deriving from our implicit knowledge depends on representing explicit knowledge in AI systems or organizations **declaratively** (as stand-alone facts or concepts, as in a database) rather than **procedurally** (as step-by-step sequences of actions, as in algorithms). The reason is that facts can be implicitly derived across many more situations than sequential procedures. For example, the fact that an omelette has eggs holds across any recipe; in contrast, even a small change to a step-by-step omelette recipe may lead to a large difference in results. How, you may ask, could we then represent inherently sequential knowledge, such as step-by-step plans? McCarthy's key idea was that any "procedural" representations should be abstracted into objects, and treated as if they were units of data. He developed a language, called **Lisp** (short for "list processing"), in which all commonsense knowledge — facts, programs, etc. — could be represented declaratively as lists.

3.2 Knowledge representations need to allow concepts at various levels of abstraction

Declarative, list-based representations should also enable representing **concepts** — that is, not just facts or programs and such things, but ways of talking about them conceptually, meaning at different **levels of abstraction**. Doing so was necessary to enable vast efficiencies for talking about specific phenomena without having to represent and reason about all the facts. For example, our implicit knowledge told us that the Pyramids of Giza should be older than my neighbor's house since we have a more abstract concept of a neighbor's house.

3.3 Knowledge representations need to reflect contexts

Consider the following sentence: "*I tried to fill the suitcase with the whale, but it was too big*". What is "it" in this sentence? We would conclude, based on our common sense, that "it" in the first one implicitly refers to the whale, since whales are usually bigger than suitcases. Yet, grammatically, "it" could just as easily refer to a whale *or* a suitcase. Imagine if I told you that the whale that was being referred to was a stuffed animal, I put it in the suitcase to take to a friend's house who had just had a baby, and that it was to be a present for their baby. In such case, the "it" in the sentence would refer instead to the suitcase!

As a second example, what a manager in an organization says or writes is embedded in goals, or what are called "**speech acts**". Consider the phrase, "Apple (the company) needs to perform better". The meaning of this sentence is about *performing some act*, which depends on *who* is saying it and for what purpose and situation — a customer may be experiencing a faulty product; an investor may be looking for Apple to improve its profits. In either case, knowledge for interpreting what is said or written cannot be handled by general common sense alone — it requires knowledge of the *pragmatic* (i.e., goal-oriented) context of the organization.

Overall, McCarthy proposed that representations of knowledge need to be specific to the **contexts** in which problems arise, where "con" means "with" the text, rather than the text itself. As in the examples above, such contexts may concern the physical reality of a situation (e.g., real vs. toy whales) or pragmatic issues such as who wants what to get done. Overall, you can think of our commonsense as existing on a kind of continuum. Some commonsense, such as the basic principles of how physical objects interact with one another, appear to hold most anywhere on earth and are believed by all people. In everyday problem-solving, however, common sense may be more reflected by the phrase "When in Rome…" — that is, facts may be true and concepts may be interpreted specific to cultural, temporal, or spatial contexts. The key takeaway is that, even if we figure out how to provide an AI system with enough explicit knowledge to derive an adequate amount of implicit knowledge, we need to enable the system also to have some common sense in how to flexibly modify its knowledge to such situation-specific contexts.

4. Elaborating commonsense knowledge

The *process* of representing knowledge in an AI system or organization — whether general commonsense knowledge or knowledge specific to the context of problems solved in a particular AI system or organization — should not be thought of in isolation. Rather, representing new knowledge should be thought of as **elaborating** a broader set of representations in a knowledge base or database. An **elaboration** refers to *adding* or *removing* units of knowledge (propositions, facts, premises, concepts, data, etc.) from the larger knowledge base or database. Further, each action in the world, or interaction (query, prompt, etc.) results in a need for novel *elaboration* of the knowledge bases or databases of an AI system. Elaborations concern operations on states. An elaboration amounts to cycling between richer and more abstract pictures of the world; even if an agent settles on a small world *act*, there is a commitment to the larger world of the underlying phenomenon.

Let us measure the complexity of knowledge representations in an AI system as the average number of elements (i.e., concepts or propositions) in a "repository" of representations of knowledge. Now imagine a measure of the number of operations for adding or removing new representations to the repository that the AI system acquires as it interacts with the world. Some operations could be needed simply to put knowledge in the "correct" format (called syntactic operations); other operations may be needed to modify which concepts or facts are valid or even relevant (called semantic operations); finally, a system may need to modify its very problem-solving goals (called pragmatic operations). All of these operations require effort and, to the extent that such effort cannot be made, think of each elaboration as leading to "garbage" piling up in the repository that is not put in the "trash" (as a simple example, a file that remains on your hard drive but has no relevance to you is semantic "garbage" that you have yet to put in your computer's trash).

McCarthy's key idea here is that having common sense knowledge requires having some automatic way of "taking out the trash" in the repository after each elaboration of it. Put another way, we have to be able to use our common sense immediately as situations unfold, such that we must have a way of flexibly modifying the knowledge in our repository, and so should an AI system (or organization). He called such a quality of an intelligent system to "take out the trash" efficiently as its **elaboration tolerance**. More formally, elaboration tolerance is defined as when *'the effort required to add new information to the repository is proportional to the complexity of that information'*. You can think of 'effort required' as the amount of 'garbage' — i.e., non-useful information — that has 'piled up' in the repository and needs to be collected.

4.1 Elaboration tolerance requires "default reasoning"

Consider our earlier sentence: "*I tried to fill the suitcase with the whale, but it was too big*". Perhaps it is a real whale, or a stuffed animal whale. Or perhaps "whale" is a nickname for some other object, such as a lunchbox with a picture of a whale on it. You can see that "whale" (or "suitcase") could in theory refer to an almost infinite number of objects and facts about the world. Hence, McCarthy proposed, elaborating our commonsense knowledge efficiently requires doing so on the basis of some *default assumptions* — our commonsense assumptions about the way the world is. In this way, we only need to change our assumptions when specific evidence arrives, such as me telling you that the whale is in fact a stuffed animal.

4.2 Elaboration tolerance requires adequately "expressive" languages

Think of the problems that SQL databases address versus a standard file system. The achieve high integrity in storing and updating data by introducing a formal language for doing so — one that is based on carefully-designed support for storying, querying and updating data in the database. In the AI idea of knowledge representation and reasoning, we call such a language as having adequate **expressiveness** — essentially, being able to express any operators you need to reason about and modify the representations of knowledge as situations change.

Think of what happens in your personal email, your hard drive, or your photo collection as you "elaborate" it, such as by adding new files, folders, photos, or messages. At least for most of us, the organization of all this information becomes something of a mess. In organizations, a similar but more intense version of this mess arises from the need for many people and departments to continually add and remove information from knowledge bases and databases that are shared and complex. In fact, the challenges of elaboration in an organization can fundamentally impair its intelligence, as reflected in how the majority of data science work is not in data-driven problem-solving per se, but rather in tasks of integrating data.

5. Approaches to representing and reasoning about knowledge

As in our previous sessions, we will understand AI ideas about knowledge representation and reasoning in terms of two paradigmatically-different approaches. One approach, which we will refer to as the **deep learning approach**, derives from the idea of Boltzmann machines that we looked at in the previous session. It views knowledge representations in an AI system as mostly emerging from patterns in bit strings. Another approach, which we will refer to as the **mathematical logic approach**, derives from McCarthy himself and views knowledge representations as emerging from patterns in lists.

5.1 Approach based on deep learning

The currently-popular approach to knowledge representation and reasoning (KRR) is based on deep learning. This approach (at least implicitly) concerns the same fundamental challenges raised by McCarthy: how to efficiently represent and reason about commonsense knowledge across situations. Yet it does so without representations that are declarative, or which can capture various concepts and contexts. Instead, knowledge is primarily represented as bit string patterns extracted from large-scale data and stored as matrix-like structures called vector databases. Knowledge such as concepts "emerge" as patterns in these bit string representations. For example, concepts are understood as statistical relationships among words or phrases in text, conjured as a measure of "conceptual distance". In large language models (LLMs), knowledge is represented through statistical patterns among tokens (bit strings that represent parts of words) and the transformations applied to them. These AI systems often now extend these bit string representations with other "external memories", such as vector databases or knowledge graphs or multimodal data (in what are called "world models", which offer an alternative deep learning-driven approach to LLMs).

To apply such representations to specific problems, modern systems rely on what is often called "context engineering." Here, relevant information is selected and provided at inference time—via prompts, retrieved documents, or other inputs—to condition the model's behavior on a particular situation. Techniques such as retrieval-augmented generation (RAG) enable systems to incorporate information from external repositories, while synthetic data can be used to approximate unobserved or hypothetical situations. In this sense, "context" is not explicitly represented within the system's knowledge base, but is instead dynamically assembled for each task.

Reasoning about situations is done through deep learning-driven statistical prediction mechanisms that are enhanced through, for example, heuristic strategies for how to prompt an LLM. Perhaps the most popular strategy, called "chain-of-thought" reasoning, is essentially to tell an LLM to reason "step-by-step". The surprisingly powerful results of this and other strategies has led proponents of the deep learning approach to think of commonsense reasoning as an "**emergent ability**", meaning that it arises spontaneously from training an LLM or other deep learning-based AI system on large volumes of data using only simple mechanisms such as statistical prediction.

These deep learning approaches have clearly been impressive. They have scaled to vast datasets and large numbers of users. They have been able to handle many contexts and generate implicit knowledge from an explicit base increasingly reliably. Yet systems developed under this deep learning approach also show fundamental limitations. First, there is little awareness or attention to the *frame problem* as the fundamental problem of credit assignment. Hence, there is little direct support for solving McCarthy's frame problem. So far at least, deep learning approaches have been used for problems where the phenomena are mostly given. They have yet to provide clear mechanisms for representing phenomena and situations precisely, nor for distinguishing what is relevant or not in a given context.

Second, though LLMs and other deep learning approaches to KRR can *accept* new information by providing new "context" (e.g., an external database) or a prompt, they do not support *elaboration* in McCarthy's sense. There is no principled way for users to add, modify, or remove knowledge such that the effects of these changes are reliably just proportional to the complexity of the new information. Updating knowledge typically involves indirect methods, such as modifying prompts, updating external data sources, or retraining the entire model, rather than ongoing and efficient operations on an evolving knowledge base.

5.2 Approach based on mathematical logic

Another approach, pioneered by John McCarthy himself, is based on *mathematical logic*. At a very basic level, mathematical logic just means using a precise, rule-based language, similar to mathematics, to describe facts and concepts about the world and to reason from them. Instead of learning statistical patterns from data and calling such patterns knowledge, this approach tries to represent such knowledge explicitly in a form that a computer can interpret and manipulate.

Knowledge is stored as clear statements about objects, properties, and relationships. For example, instead of implicitly learning that "pyramids are old," a system might explicitly represent a statement like: "X is older than Y." The goal is to make knowledge transparent and precise, so that the system can reason about it step by step. This is done through formal languages for expressing knowledge that an AI system can both represent and reason with. To apply such representations to specific problems, AI systems in this approach construct rich conceptual structures, such as formal databases and **ontologies** — categorizations of phenomena into objects. As a noted recent example, the software platform Palantir builds vast ontologies that structure data for specific client organizations in terms of real-world "entities" (such as people, organizations, and events) and their relationships, allowing these organizations to precisely represent and reason about complex, evolving situations in ways that are relevant to their decision-making processes.

AI systems in a mathematical logic approach also use explicit reasoning procedures (rather than heuristic strategies such as "chain-of-thought") to draw inferences from these ontologies. For instance, reasoning procedures could start from known and explicitly-represented facts, then apply rules to derive new facts; alternatively, they may be given a goal state represented explicitly in a formal language and work backward to infer what must be true in a situation. Reasoning under this approach is not an "emergent property", but rather follows from the logical reasoning procedures. A clear advantage is that this formalization means that it is always possible to reason precisely about how any conclusions are reached, and to modify knowledge in a principle fashion. Hence, this approach more directly supports elaboration tolerance, where knowledge can be formalized and added or removed in a controlled manner.

These logic-based approaches have clear strengths. They offer precision, interpretability, and control. They directly address issues like the frame problem by trying to formally specify what changes and what stays the same across situations. In domains where correctness and clarity matter—such as planning, scientific reasoning, or structured decision-making—these properties are especially valuable.

At the same time, they have limitations, at least for now. Instead of drawing on patterns that emerge spontaneously from data, the aim is to build intelligence from logically-precise representations and carefully defined reasoning processes for learning these. At least at some level, a requirement seems to be to deliberately represent at least a good chunk of commonsense knowledge for such systems. Hence, though such an AI system's knowledge should be easier to track and understand, it is harder for now to construct such systems at scale.